

LESSON 5-1 • SESSION 1 • GUIDED NOTES**Area of Parallelograms**

Learning target – I can find the area of a parallelogram by multiplying the base by the perpendicular height. · 6.G.A.1

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
parallelogram <i>paralelogramo</i>	A four-sided shape whose opposite sides are _____.	opposite sides never meet
base <i>base</i>	The side you _____ the height from.	b in $A = b \times h$
height <i>altura</i>	The distance from the base to the opposite side, at a _____ corner.	h in $A = b \times h$

KEY IDEA*complete each sentence*

- Area of a parallelogram: $A = \underline{\hspace{1cm}} \times \text{height}$.
- The height makes a _____ corner with the base. The slanted side is _____ the height.
- Missing height or base: _____ the area by the measure you know.

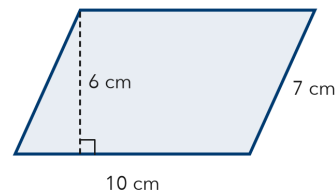
WORKED EXAMPLE*fill in the blanks as we solve*

A parallelogram has a base of 9 in., a height of 4 in., and a slanted side of 5 in. Find its area.

Step 1 base = _____ in., height = _____ in.**Step 2** Skip the 5 in. slanted side.**Step 3** $A = 9 \times 4 = \underline{\hspace{1cm}}$ **Answer** The area is _____ in².**YOU TRY***use the example as a model*

1. Find the area.

Show your work.

**Answer** _____ cm²

2. Find the missing measure.

a. $A = 42 \text{ in}^2$, $b = 7 \text{ in}$, $h = \underline{\hspace{1cm}} \text{ in}$

b. $A = 54 \text{ m}^2$, $h = 6 \text{ m}$, $b = \underline{\hspace{1cm}} \text{ m}$

SUM IT UP*in your own words*

I do not use the slanted side to find area because ____.

LESSON 5-1 • SESSION 2 • GUIDED NOTES

Area of Rhombuses

Learning target – I can find the area of a rhombus using the base and height or the two diagonals. · 6.G.A.1

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
rhombus <i>rombo</i>	A parallelogram with four sides of the ____ length.	all 4 sides equal
diagonal <i>diagonal</i>	A segment from one corner to the ____ corner.	a rhombus has 2 diagonals
congruent <i>congruente</i>	Exactly the same ____ and shape.	4 congruent sides

KEY IDEA*complete each sentence*

- A rhombus is a _____, so $A = \text{base} \times \text{height}$ works.
- With the two diagonals, $A = d1 \times d2 \div \underline{\hspace{1cm}}$.
- Missing diagonal: _____ the area, then divide by the diagonal you know.

WORKED EXAMPLE*fill in the blanks as we solve*

A rhombus has diagonals of 10 cm and 8 cm. Find its area.

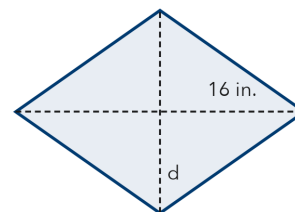
Step 1 Two diagonals are given, so use $A = d1 \times d2 \div 2$.**Step 2** $10 \times 8 = \underline{\hspace{1cm}}$ **Step 3** $80 \div 2 = \underline{\hspace{1cm}}$ **Answer** The area is $\underline{\hspace{1cm}}$ cm^2 .**YOU TRY***use the example as a model*

1. Find the area of each rhombus.

a. diagonals 6 m and 12 m _____ m^2 **b.** base 7 cm, height 5 cm _____ cm^2

2. The rhombus has an area of
- 72 in^2
- . Find the missing diagonal
- d
- .

Show your work.

Answer _____ in.**SUM IT UP***in your own words*

When I use the diagonals, I divide by 2 because ____.
